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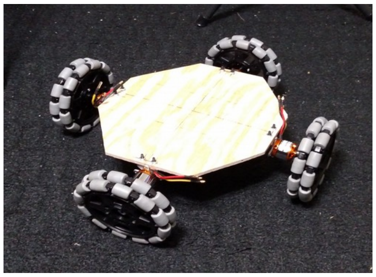
Weekly Report 7 due February 23, 2016

Welp, here we are. It's been quite a while since my last report – unfortunately I've been pretty busy with outside obligations and have only now been making progress. I've been sprinting to catch up with my peers and make this Thursday's object avoidance demo deadline, so here's hoping I'll make it!

Anyways, some things that I have to report on:

- I haven't really done anything more with my algorithm that I'll be using in the final catching robot. Sorry!
- I've successfully interfaced ultrasonic sensors with my TI launchpad and had basically no problems.
- I had a bit of trouble trying to get my motors to work with my wheels. My initial plan was to use set screws on my wheel hubs that would connect to my hex adapters (which are connected to my motor shaft), but as the TA of the student machine shop pointed out, there wasn't quite enough wall thickness to recommend this plan.
    - Currently my work around is to use tape. Yup. Tape. Will it fail? Maybe…? Only time will tell!
    - I plan to use my home equipment over Spring Break to come up with a more permanent solution
- In the spirit of planning before doing, I still haven't cut out my chassis. The temporary setup that I'm using for object avoidance is shown below:



        - She ain't pretty, but she'll hold together!
            - You know… hopefully

Some things that I've definitely got to do after this object avoidance demo is out of the way:

- Finalize the chassis design and cut it out
- Actually design the ball launching mechanism…
- Actually use the camera with the Jetson to track and react to the thrown balls.
- Use the Wi-Fi modules on the TI launchpads to talk between the launcher and catcher mechanism.

All in all, while I've been pre-occupied up until now, I plan on working extra hard during Spring break to catch up. Though it may be a personality flaw, I stubbornly refuse to give up on my original vision for the ball-catching robot, and am determined to get it working in a moderately impressive manner. Here's hoping that things don't blow up in my face!